GG-C-136b
SEPTEMBER 4, 1958

SUPERSEDING
Int. Fed. Spec. GG-C-00136a (GSA-FSS)
June 11, 1957 and
Fed. Spec. GG-C-136
December 9, 1947

FEDERAL SPECIFICATION

CATHETERS, URETERAL, (X-RAY, GRADUATED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers ureteral catheters X-Ray graduated in three types and six sizes.

1.2 Classification.

1.2.1 *Types and sizes.*—Graduated X-Ray ureteral catheters shall be of the following types and sizes, as specified.

Type I.—Olivary tip
Type II.—Round tip
Type III.—Whistle tip
Size 4—4 French
Size 5—5 French
Size 6—6 French
Size 7—7 French
Size 8—8 French
Size 10—10 French

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-B-10377—Box, Wood, Cleated, Veneer, Paper, Overlaid.

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MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Woven fabric.* — Unless otherwise specified, the fabric shall be woven from good quality nylon, silk, or rayon yarn, of a size appropriate for the diameter of the catheter.

3.1.2 *Coating.* — The coating shall be a flexible varnish, or equal, containing lead or other X-Ray opaque material.

3.2 General description.—Graduated X-Ray ureteral catheters shall each be composed of a tightly and smoothly woven base, free from loose ends or broken fibers, and completely, uniformly, and symmetrically covered with a flexible coating containing the X-Ray opaque material. The finished catheter shall have the proper flexibility, pliability, and body required to withstand the pressure of introduction through the cystoscope and into the ureter, together with the softness and pliability necessary to prevent irritation to body tissues. Catheters shall be 27 to 28 inches (68.5 to 71 cm.) in over-all length. Catheter diameter shall be determined as specified in 4.3.2. The catheter being checked, shall pass freely through the specific scale opening of the size designated, but not through the next smaller size opening of the French scale.

3.3 Base.—The base shall be a single web, tightly and smoothly woven in braided form with yarn of appropriate size and type for the size of the tube in question on a 26 to 40 carrier machine. The thickness of the wall shall be relatively uniform throughout the length of the tube. Each yarn woven into the tube shall be of one continuous length from one end of the tube to the other. The tube shall have the required body to secure resistance to buckling or breaking when tested in accordance with 4.3.1, 4.3.3, 4.3.4, and 4.3.5.

3.3.1 *Eyes.*—Two eyes shall be smoothly and symmetrically woven into the structure of the tube on opposite sides of the tube. The area of each eye shall be not less than the cross-sectional area of the catheter. They shall not overlap nor be separated by more than 1 centimeter.

3.4 Coating.—The coating shall thoroughly and uniformly impregnate in and around the woven base in a manner to coat the braided fibers completely. The coating thickness shall be relatively uniform throughout the length of the catheter. The coating shall be formulated to secure maximum adhesion to the tube and shall have the required toughness and elasticity to permit repeated flexing and to prevent peeling, cracking, or breaking when tested in accordance with 4.3.3, 4.3.4, and 4.3.5. The coating shall be resistant to body fluids, mild acids, alkalies, and deterioration due to aging during storage. The coating catheter shall show no signs of brittleness, blistering, or other visible damage as a result of the boiling or auto-claving tests described in 4.3.3 and 4.3.4. The coated catheter shall not be tacky, shall not stick together, and its surface shall not be marred nor show any other evidence of damage as a result of the flexibility tests described in 4.3.5.

3.5 Toxicity.—The catheters shall be free from toxic material or any ingredient that might have a deleterious effect upon body tissues.

3.6 X-Ray opacity. — The catheters shall throw a definite, uniform shadow, when

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tested as described in 4.3.6. The degree of opaqueness shall be such that the shadow of a small foreign body in the front or rear of the catheter shall be visible for diagnosis.

3.7 **Finish.**—The surface of each catheter shall be properly worked down and given a dry, smooth, highly polished but elastic finish, free from scratches, grooves, or other defects which would prevent its easy passage through the cystoscope or the ureter. The surface shall be "slick" when wet and shall pass easily.

3.8 **Stylet.** — Each catheter shall be furnished with a stylet of stainless-steel wire of proper size to pass freely through the full length of the lumen.

3.9 **Marking.**—Each catheter shall be plainly and indelibly marked upon the proximal end with the manufacturer's name, its size according to the French scale, and the word "X-Ray". Each catheter shall be graduated in the metric system, based upon groupings of 1, 5, and 25 centimeters. These graduations shall be legibly and indelibly marked.

3.10 **Type I, olivary tip.**—The tip of the type I catheter beyond the distal eye shall be formed or molded of the same material as the catheter itself. The tip shall consist of a conical point surmounted by a small spheroidal enlargement. The construction shall be such as not to weaken the structure or materially reduce the flexibility of the catheter. The catheter shall be slightly tapered between the last eye and the bulbous tip.

3.11 **Type II, round tip.**—The tip of the type II catheter beyond the distal eye shall have a rounded end and shall be of the same size as the remainder of the catheter.

3.12 **Type III, whistle tip.**—The tip of the type III catheter shall be formed by placing a third opening beyond the distal eye at the extreme tip, as in a whistle.

3.13 **Sizes.**—Catheters shall be of the following sizes, and each catheter shall be woven on a mandrel of the applicable diameter shown in table I. The lumen of each catheter shall be such that, after soaking, the catheter will be capable of passing freely a mandrel of slightly smaller diameter, as specified in 4.3.1.

TABLE I.—*Mandrel diameters for weaving catheters*

Catheter size	Mandrel size
	Inch
4 French	0.020
5 French	0.029
6 French	0.039
7 French	0.045
8 French	0.051
10 French	0.070

3.14 **Workmanship.** — The workmanship shall be first class throughout. The finished catheters shall be free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**

4.1.1 *For visual examination.*—Sampling procedures for visual examination shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.1.2 *For destructive tests.*—Sample size for each test shall be as indicated for inspection level L2 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero (0) for all sample sizes, and a minimum of two (2) samples shall be taken regardless of lot size.

4.1.3 *For X-Ray opacity test.*—Sample size for test shall be as indicated for inspection level L1 in Appendix to Military Standard MIL–STD–105, except that the acceptance number shall be zero for all sample sizes, and a minimum of two samples shall be taken regardless of lot size.

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4.2 **Inspection.**—The inspector shall ascertain that the catheters have been examined and tested in accordance with specification requirements.

4.3 **Tests.**

4.3.1 *Mandrel test.*—Each sample catheter shall be immersed for 3 hours in distilled water at room temperature. The applicable mandrel shown in table II shall then be passed completely through the catheter to the eye nearest the tip.

4.3.2 *Outside diameter.*—The outside diameter shall be tested for French size against a standard French scale (where one French is equal to 1/3 millimeter).

4.3.3 *Boiling test.*—Each sample catheter shall be boiled for ten 20-minute intervals, with 24 hours between each boiling to permit drying out at room temperature (68° to 72°F.).

4.3.4 *Autoclaving test.* — Each sample catheter selected for test shall be dry-autoclaved for ten 10-minute intervals at 250°F., with sufficient time between to permit cooling to room temperature (68° to 72°F.).

4.3.5 *Flexibility tests.*

4.3.5.1 *Wet test.*—The catheter shall be twisted three full length turns around a catheter of the same size, subjected to boiling for 20 minutes, allowed to return to room temperature, then untwisted.

4.3.5.2 *Dry test.*—The catheter shall be twisted three full length turns around a catheter of the same size, subjected to autoclaving for 15 minutes at 250°F., allowed to return to room temperature, then untwisted.

4.3.6 *X-Ray opacity.*—Each sample catheter shall be radio-graphed as follows:

85 KVP, 10 milliampere-seconds, 25-inch distance, 24 centimeters of U.S.P. paraffin superimposed over catheters, and using a standard make of cassette with screens, films, and processing solutions.

The foreign body used for test purposes shall be a small fragment of sheet lead 1/2 millimeter in thickness.

TABLE II.—*Mandrel diameters for testing catheters*

Catheter size	Mandrel size	
	Minimum	Maximum
	Inch	*Inch*
4 French	0.016	0.018
5 French	.024	.026
6 French	.033	.036
7 French	.038	.041
8 French	.044	.047
10 French	.063	.065

4.4 **Supplier responsibilities for inspection.**—The supplier shall perform examinations and tests set forth in this specification to substantiate conformance of supplies to specification requirements. The examinations and tests shall be at the expense of the supplier. Suppliers shall have adequate test facilities or shall arrange for test facilities acceptable to the Government to be used for executing specific prescribed tests. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time or from time to time, for a period of 2 years after the delivery of the supplies to which these records relate. The supplier shall be responsible for compliance with all requirements of this specification, and all other technical requirements of the contract.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing protection used are defined in Fed. Std. 102.

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5.1 Packaging.

5.1.1 *Unit of issue.*—One catheter, as specified, constitutes one unit.

5.1.2 *Packaging quantities.*—The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE III.—*Packaging quantities*

Unit package	Intermediate package	Shipping container
1 unit ----------	6 units	360 units

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each catheter shall either be packaged (not coiled) in an envelope fabricated from polyethylene, or waxed, or glassine paper, and secured, or shall be completely wrapped (not coiled) in waxed or glassine paper and secured.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 *Level C.*—Unless otherwise specified, commercial packages are acceptable.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind, commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Interior packages.*—Each unit package and each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

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6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title number and date of this specification.

(*b*) Types and sizes required (see 1.2.1).

(*c*) Fabric required (see 3.1.1).

(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Armed Services Medical Stock List:

6515–316–5200 CATHETER, URETERAL, X-Ray, Olive Tip, 4 Fr.
6515–316–5210 CATHETER, URETERAL, X-Ray, Olive Tip, 5 Fr.
6515–316–5220 CATHETER, URETERAL, X-Ray, Olive Tip, 6 Fr.
6515–316–5230 CATHETER, URETERAL, X-Ray, Olive Tip, 7 Fr.
6515–316–5240 CATHETER, URETERAL, X-Ray, Olive Tip, 8 Fr.
6515–316–5250 CATHETER, URETERAL, X-Ray, Olive Tip, 10 Fr.
6515–316–6200 CATHETER, URETERAL, X-Ray, Round Tip, 5 Fr.
6515–316–6210 CATHETER, URETERAL, X-Ray, Round Tip, 6 Fr.
6515–316–6220 CATHETER, URETERAL, X-Ray, Round Tip, 7 Fr.
6515–316–6800 CATHETER, URETERAL, X-Ray, Whistle Tip, 4 Fr.
6515–316–6810 CATHETER, URETERAL, X-Ray, Whistle Tip, 5 Fr.
6515–316–6820 CATHETER, URETERAL, X-Ray, Whistle Tip, 6 Fr.
6515–316–6830 CATHETER, URETERAL, X-Ray, Whistle Tip, 7 Fr.
6515–316–6840 CATHETER, URETERAL, X-Ray, Whistle Tip, 8 Fr.
6515–316–6850 CATHETER, URETERAL, X-Ray, Whistle Tip, 10 Fr.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 Transportation description.—Transportation description applicable to this item is:

Surgical instruments, not otherwise indexed by name.
Carload minimum weight 10,000 pounds, subject to Rule 34, Uniform Freight Classification.
Motor volume minimum weight 10,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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